

Deep research on your SaaS future thesis

Core thesis and how it differs from the “agents replace SaaS” narrative

Your memo argues for a middle path between two popular extremes: (a) “agents will replace most people and collapse software teams into one person with bots,” and (b) “GenAI will let every company vibe-code everything internally, so SaaS becomes obsolete.” Instead, you propose that GenAI shifts the *scaling function* of software: fewer people can build more, which expands the feasible set of tools that can exist, but it does **not** remove the economic value of “productisation” (reliability, security, governance, operations, support, compliance). This framing aligns with what early empirical research suggests: AI coding tools can materially increase individual throughput, but production-grade software still requires socio-technical coordination, operational discipline, and risk management that do not disappear just because code is cheaper to produce. ¹

A key implication of your thesis is that the SaaS category does not vanish; it **moves up the stack**. What customers increasingly pay for is not “someone typed the code,” but “this capability is continuously correct enough, safe enough, observable enough, and legally/commercially supportable enough to be depended upon.” That emphasis is consistent with widely used operational doctrines (e.g., SLOs and error budgets) that treat reliability as an explicit product decision rather than an afterthought. ²

Finally, your memo makes a market-structure claim: when software creation becomes radically cheaper, the long tail of specialised tools becomes economically viable, creating a “fat middle class” of small-to-mid-sized software companies (often open-source-led) while weakening some incumbent rents—especially where switching costs and lock-in have historically been high. Contemporary discussions of platform “enshittification” explicitly connect quality decline to lock-in and high switching costs, which provides a vocabulary for the “rent extraction vs customer value” dynamic you referenced. ³

What the data says about agentic development and individual leverage

The strongest near-term evidence for “one person can do more” comes from studies of AI coding assistants: controlled experiments and field trials repeatedly find meaningful speed or throughput gains, with variation by skill level and task type.

A widely cited controlled experiment on an engineering task (implementing an HTTP server) found that developers given access to an AI pair programmer completed the task **55.8% faster** than a control group. ⁴

However, the more probative question for your thesis is: do these gains persist in real organisational settings, on real codebases, with real constraints? A large set of workplace RCTs across three enterprises reported that—while each individual experiment was noisy—pooling results across **4,867 developers** showed an estimated **26.08% increase** (standard error 10.3%) in completed tasks among developers using the tool, with larger gains and higher adoption among less experienced developers. ⁵

This matters for your “teams of 10 do the work of 100” analogy: the empirical results are not yet “10× across the board,” but they are consistent with a world where *effective capacity expands* and where bottlenecks reallocate away from raw code entry. This is reinforced by adoption expectations from analysts: [Entity] [“company”, “Gartner”, “research advisory firm”] has projected rapid uptake of AI code assistants in enterprises (for example, moving from “<10% using” to a large majority within a few years).

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Agentic coding capability is also measurable in benchmark progress. For example, the SWE-bench project’s official leaderboards show agentic systems climbing to high success rates on “Verified” tasks, indicating that multi-step issue resolution and patch generation is improving quickly. 7 The important nuance for your paper is that “high benchmark performance” is *not identical* to “production reliability”: even strong benchmark results imply a non-trivial failure rate, and the remaining failures cluster around exactly the kinds of messy edge cases that dominate mission-critical operations (ambiguous requirements, partial observability, integration boundaries, security constraints, and organisational accountability). 8

Your memo also claims the true historical bottleneck was not “typing code,” but coordination: alignment on user goals, refactoring and simplification, and the overhead of integrating data and systems. Classic software-engineering management work formalised why adding people can slow delivery: communication overhead, ramp-up time, and task interdependence. That mechanism does not disappear when AI increases individual output; if anything, it can become *more* binding because throughput increases upstream. 9

Why “vibe-coded” internal tools will explode, but won’t eliminate SaaS

Your memo’s “Excel is already internal software” point is well supported by the end-user development literature: spreadsheets are widely treated as a dominant end-user programming environment, and academic work has explicitly argued that spreadsheet formula languages function as a de facto mass programming platform. 10

What GenAI changes is the slope of the curve: more employees can create more software artefacts (scripts, automations, connectors, dashboards, small apps) with less training. This is consistent with the broader “citizen developer” and low-code/no-code trajectory: [Entity] [“company”, “Gartner”, “research advisory firm”] has predicted that a very large share of new applications would be developed using low-code/no-code technologies, and it defines a “citizen developer” as an employee creating application capabilities outside traditional IT reporting lines using tools not actively forbidden. 11

So the “internal tool explosion” in your memo is not speculative—it is a continuation of a long-running pattern, now amplified by LLM-based assistance. The counterweight, which your thesis emphasises, is that this also expands **risk surfaces**: unreviewed automations, unclear data handling, and invisible dependency chains. The security community has long described this as “shadow IT,” and the [Entity] [“organization”, “National Cyber Security Centre”, “uk cyber security agency”] warns that unsanctioned or unknown assets create governance and security exposure. 12

This creates a plausible structural result that supports your “SaaS resurgence” claim: organisations will build far more **prototypes and local optimisations**, but as soon as a tool becomes business-critical, they will either (a) professionalise it internally through platform engineering, or (b) externalise it to a vendor whose product is designed to run reliably with operational guarantees.

Notably, “platform engineering” is itself an industry response to the “companies don’t invest in productivity tooling” observation you made. Analyst guidance frames platform engineering teams as internal providers of reusable services and tools that increase delivery throughput; one forecast expects a large majority of big engineering organisations to establish such teams within a short horizon. ¹³ This is essentially the formalisation of “make others more productive” as a first-class organisational function—which matches your assumption that the bottleneck historically sat in tooling, integration, and coordination rather than in raw coding ability.

The productisation gap: why reliability, security, and governance become the differentiator

Your memo’s central mechanism—“anyone can prototype, but productisation is hard”—is strongly defensible, and it can be made concrete by breaking “productisation” into measurable disciplines.

Reliability engineering is one component. The SRE framing (service level objectives, error budgets, and written policies that trade off release velocity against reliability) exists precisely because mission-critical systems cannot be “mostly correct most of the time.” Reliability is treated as a managed product constraint, not just a technical detail. ²

Security is another component, and GenAI/agentic systems introduce distinctive failure modes that support your argument that “flaky” or “hallucination-prone” behaviour is unacceptable in critical workflows. The `Entity["organization","National Cyber Security Centre","uk cyber security agency"]` has argued that prompt injection is not analogous to SQL injection because current LLMs do not enforce a robust boundary between “instructions” and “data” inside prompts. This is highly aligned with your claim that language-model behaviour includes an interpretive layer that is sometimes the opposite of what you want in deterministic systems. ¹⁴

Similarly, the `Entity["organization","OWASP","web security nonprofit"]` “Top 10 for Large Language Model Applications” catalogues common risks (prompt injection, insecure output handling, training data poisoning, and others) and treats them as a security engineering problem, not a temporary novelty. ¹⁵

Agentic systems can also create *compositional* risks: when multiple agents can call each other or act with delegated privileges, injection or confusion can escalate. Public reporting on second-order prompt injection in enterprise agent setups (where untrusted content influences an agent, which then recruits higher-privileged agents) demonstrates how operational configuration, privilege boundaries, and supervision loops become essential parts of “productisation.” ¹⁶

Compliance and auditability are a third component. Many enterprise buyers treat SOC 2-style assurance as a standard procurement gate: the `Entity["organization","AICPA","us accounting standards body"]` describes SOC 2 examinations as reports on controls relevant to security, availability, processing integrity, confidentiality, and privacy—attributes that map cleanly onto your “it must run all the time and be governable” definition of productisation. ¹⁷ Similarly, `Entity["organization","ISO","international standards body"]` positions ISO/IEC 27001 around implementing an information security management system (ISMS), again reflecting that “being depended upon” is as much about process and control as it is about software features. ¹⁸

Your memo also gestures at the economics of GenAI operation—“running inference is expensive.” That claim is increasingly supported by energy system analysis: the `Entity["organization","International Energy Agency","intergov energy org"]` projects that global electricity consumption for data centres

could roughly double to around 945 TWh by 2030 in its base case, reaching just under 3% of global electricity consumption, with rapid growth driven in part by AI workloads. ¹⁹ Even if model efficiency improves, this macro constraint reinforces your argument that turnkey, well-governed delivery (including cost controls, caching, evaluation, and workload shaping) becomes part of what customers pay for—again nudging value towards professional SaaS and managed offerings rather than ad hoc internal deployments. ²⁰

SaaS industry restructuring: from monoliths and rents to a “fat middle” of specialised vendors

Your “big SaaS companies will be shocked, and the mid-market of specialist vendors grows” thesis can be grounded in three interacting forces: lower build cost, higher switching feasibility, and demand for domain-specific “last-mile” fit.

Lower build cost is reasonably evidenced by developer tooling studies covered above: when tasks become 25–55% faster in many contexts, the addressable set of “economically buildable” applications expands, especially for narrow workflow problems. ²¹ The likely result is not “no SaaS,” but more SaaS categories and more entrants—particularly vertical and micro-SaaS products designed around specific functions, as you described with finance, budgeting, expenses, and procurement controls.

Switching feasibility is partly policy-driven, and this is where your Europe-centric optimism can be sharpened. The `Entity["organization","European Commission","eu executive body"]` describes the EU AI Act’s staged applicability timeline (with obligations phasing in before full applicability), which increases compliance stakes for AI-driven products and can indirectly advantage vendors that professionalise governance and documentation. ²² Separately, the EU Data Act explicitly targets barriers to cloud switching and portability, becoming applicable (including for key switching provisions for “data processing services”) from September 12, 2025—supporting your claim that more interoperability and portability can reduce lock-in and increase customer options. ²³

Demand for domain-specific fit is a core part of your memo (“customers used to buy 100% features to get 10% of what they need”). While the exact “10%” number varies by context, the existence of SaaS waste and underutilisation is widely documented in practice, including public audits that estimate material costs from unused licences in large organisations. This pattern supports your argument that buyers often overpay for broad suites when their actual needs are narrower and more contextual. ²⁴

Your reference to “enshittification” (quality degradation driven by extraction incentives) adds a market-structure hypothesis: once customers have credible alternatives (via lower build cost, open-source options, and better switching), vendors that rely on lock-in and rent extraction lose pricing power unless they reinvest in customer value. Mainstream coverage describes “enshittification” as a lifecycle where platforms shift value away from users and partners toward shareholders, often enabled by lock-in. ²⁵

This is where your open-source claim becomes more than ideology. Enterprise open-source adoption is widespread across core stack layers, and recent survey-based reporting shows meaningful penetration across operating systems, cloud/container tooling, web/app development, data management, and AI/ML. ²⁶ Meanwhile, public-sector and EU-level strategies explicitly encourage reuse and “Think Open” practices, signalling that (at least in Europe) procurement and governance can push toward open components and interoperability rather than closed suites. ²⁷

Putting these together yields a coherent “fat middle” forecast consistent with your memo: many small-to-mid-sized vendors build and maintain focused, high-quality tools (often open-source-core or open-

standards-aligned), while a smaller set of large platforms remain as infrastructure layers and distribution channels—but face more competitive pressure on pricing and quality than in a high lock-in environment. ²⁸

Employment and capability implications: more builders, different roles, new failure modes

Your memo claims “we will need more developers, not fewer,” because demand for software will expand as creation gets easier, and because every business function becomes more tool-producing. Labour market evidence is mixed and time-dependent, but several reputable projections still expect growth in software development roles even under automation pressure.

In the `Entity["country","United States","sovereign state"]`, the `Entity["organization","Bureau of Labor Statistics","us labour stats agency"]` projects employment for software developers, QA analysts, and testers to grow 15% from 2024 to 2034, with large annual openings driven by both growth and replacement needs. ²⁹ Globally, the `Entity["organization","World Economic Forum","international ngo"]` Future of Jobs reporting lists software and application developers among the fastest-growing roles in percentage terms, alongside AI/ML specialists and other tech roles. ³⁰ In the `Entity["country","United Kingdom","sovereign state"]`, government-commissioned analysis projects substantial growth in jobs directly involving AI activities over the next decade, implying expansion in adjacent roles that build, integrate, and govern these systems. ³¹

At the same time, your paper will be stronger if it acknowledges two countervailing forces that show up in the data:

First, job categories can bifurcate. Reporting on labour statistics has highlighted that narrower “programmer” roles (implementation-heavy) can contract even as broader “software developer” roles remain resilient—consistent with your argument that the value shifts toward orchestration, architecture, domain alignment, and risk management rather than raw coding. ³²

Second, “more internal tooling” without governance can increase security and compliance incidents. The shift from shadow IT to “shadow AI” is already a concern in cyber guidance, and the UK cyber community has warned that GenAI can amplify phishing and integrity risks, increasing the organisational need for controlled tooling and specialist oversight. ³³

The upshot is aligned with your memo, but it should be stated precisely: the total volume of “software-building work” is likely to increase, while the content of professional software roles shifts toward (a) system design and integration, (b) quality and reliability engineering, (c) security threat modelling and control design for agentic systems, and (d) domain-expert product development that captures the “last 20%” of contextual fit. ³⁴

A defensible model you can use in your paper and testable predictions

To turn your memo into a rigorous paper, it helps to formalise your argument as a model with observable implications.

A compact model consistent with the evidence is:

1) GenAI reduces the cost of producing first-draft code and accelerates prototyping and iteration; 2) as prototype supply increases, organisational bottlenecks shift to coordination, integration, and governance; 3) the value of SaaS shifts from “feature inventory” to “operationalised capability,” with reliability/security/compliance as differentiators; 4) market structure fragments into many specialised vendors plus a smaller set of platform layers; 5) policy (interoperability, portability, AI governance) increases customer switching power, especially in jurisdictions prioritising these goals. ³⁵

From that model, you can propose predictions that are both falsifiable and measurable:

A prediction about internal build: enterprises will ship substantially more internal apps and automations (often via low-code/no-code + GenAI), but will also expand governance functions to control shadow IT and agentic risk; measurable via growth in platform engineering adoption, internal tool counts, and security policy enforcement. ³⁶

A prediction about SaaS differentiation: SaaS winners will compete less on “checkbox features” and more on demonstrable operational excellence (SLO transparency, audit readiness, secure-by-default agent configurations, and cost governance for inference); measurable via procurement requirements (SOC 2 / ISO27001), published reliability commitments, and buyer insistence on AI/agent governance controls. ³⁷

A prediction about vendor landscape shape: the number of viable niche/vertical vendors increases because AI lowers go-to-market and build costs, while “suite” incumbents face pressure in segments where switching becomes easier (e.g., via Data Act portability rules in Europe); measurable via vendor counts in specific vertical categories, churn rates, and the prevalence of data-export/switching features as procurement criteria. ³⁸

A prediction about open source economics: open-source-led products gain share where buyers require sovereignty, interoperability, and controllability; measurable via enterprise open source adoption surveys and public-sector procurement patterns. ³⁹

A prediction about labour: “software developer” and adjacent roles (AI governance, platform engineering, security engineering for agentic systems) grow even if narrow “programmer” roles compress; measurable via occupational employment splits and hiring trends, plus training and skills initiatives. ⁴⁰

A prediction about cost constraints: energy and compute constraints become a strategic limiter on indiscriminate agent deployment, pushing vendors toward efficiency, smaller/specialised models, and hybrid architectures; measurable via inference cost management practices and data centre energy constraints discussed in energy-system analyses. ²⁰

¹ ⁵ ²¹ demirermert.github.io

https://demirermert.github.io/Papers/Demirer_AI_productivity.pdf

² ⁸ Defining slo: service level objective meaning

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³ ²⁵ The Age of Enshittification

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⁴ ³⁵ The Impact of AI on Developer Productivity: Evidence from GitHub Copilot

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